Quantum Neural Networks and Quantum Physics-Informed Neural Networks

From Input Encoding to PDE Solvers

Morten Hjorth-Jensen

Department of Physics, University of Oslo

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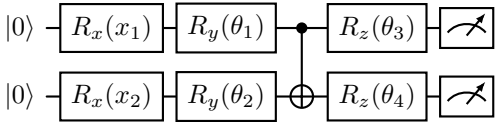
Outline I

Outline II

QNN as a Variational Quantum State

Quantum Neural Network (QNN):

$$|\psi(x,\theta)\rangle = U(\theta)\,U(x)\,|0\rangle^{\otimes n}$$
$$f(x,\theta) = \langle\psi|\,O\,|\psi\rangle$$



- $U(x)$: **feature map** — encodes classical input
- $U(\theta)$: **variational ansatz** — trainable
- O: **observable** — defines the output scalar

Pauli rotation ansatz

$U(\theta) = \prod_\ell e^{-i\theta_\ell P_\ell/2}$, $P_\ell \in \{X,Y,Z\}^{\otimes n}$

Parameter-shift rule

$\partial_{\theta_j} f = \frac{1}{2}[f(\theta_j+\frac{\pi}{2}) - f(\theta_j-\frac{\pi}{2})]$. Exact; 2 evals per param.

QFI, TDVP, and Barren Plateaus: Quick Recap

Quantum Fisher Information (QFI):

$$F_{jk} = 4\,\mathrm{Re}\big(\langle\partial_j\psi|\partial_k\psi\rangle - \langle\partial_j\psi|\psi\rangle\langle\psi|\partial_k\psi\rangle\big)$$

Fubini–Study metric on the variational manifold.

TDVP: $\sum_k F_{jk}\dot{\theta}_k = \mathrm{Im}\langle\partial_j\psi|H|\psi\rangle$
Natural gradient: $\dot{\boldsymbol{\theta}} = -\mathbf{F}^{-1}\nabla_\theta\mathcal{L}$

BCH and depth–expressivity

$U^\dagger O U = O + i[H_\theta, O] + \frac{(i)^2}{2!}[H_\theta,[H_\theta,O]] + \cdots$
Depth $\leftrightarrow$ hierarchy of correlations.

Barren plateaus

Unitary 2-designs give $\mathrm{Var}\big(\partial_{\theta_j}\mathcal{L}\big) \sim 1/2^n \to 0$. Mitigation: physics-inspired *shallow* ansätze — precisely what QPINNs exploit.

From Discriminative QNNs to Physics-Informed QNNs

Discriminative QNN

- Input: labelled data pair (x_k, y_k)
- Loss: $\mathcal{L} = \sum_k (f(x_k;\theta) - y_k)^2$
- Learns a mapping from observations
- No physical constraint enforced

Physics-informed QNN (QPINN)

- Input: collocation points $x_k \in \Omega$
- Loss: $\mathcal{L} = \|\mathcal{D}[f_\theta]\|^2 + \lambda\|\mathcal{B}[f_\theta]\|^2$
- Learns the *solution* of the PDE
- Physics is embedded in the loss function

Why quantum circuits for PDEs?

1. Exponential compression via entanglement
2. Natural wave-like encoding via rotation gates
3. Input-shift rule: spatial derivatives at zero extra circuit depth
4. Connection to quantum simulation of differential operators
5. QFI/TDVP: optimal optimisation geometry

Why Encoding Matters

A variational quantum circuit takes classical data $\mathbf{x} \in \mathbb{R}^d$ and produces a quantum state. The **feature map**

$$\Phi : \mathbb{R}^d \longrightarrow \mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}, \qquad \mathbf{x} \mapsto |\phi(\mathbf{x})\rangle,$$

determines:

- what functions the circuit can represent (expressivity),
- the induced kernel on the data space,
- trainability (gradient norms depend on the map).

Principle

The encoding is part of the model. Choosing it well is as important as designing the variational ansatz. For PDEs, the encoding also determines which spatial frequencies the circuit can resolve.

Basis Encoding

Idea. Map a binary string $\mathbf{x} \in \{0,1\}^n$ to the computational basis state:

$$\mathbf{x} = (x_1, \ldots, x_n) \longmapsto |x_1 x_2 \cdots x_n\rangle.$$

Procedure.

1. Start in $|0\rangle^{\otimes n}$.
2. Apply X_i (NOT gate) on qubit i wherever $x_i = 1$.

Properties.

- Only n gates; depth $O(1)$.
- Requires n qubits for n bits.
- Superposition: encode N strings using $\lceil \log_2 N \rceil$ qubits *if* state preparation is efficient.
- Not suited to continuous features or PDEs.

Angle Encoding

Idea. Encode $x_i \in \mathbb{R}$ into the rotation angle of qubit i:

$$|\phi(\mathbf{x})\rangle = \bigotimes_{i=1}^{n} R_y(\pi x_i)|0\rangle = \bigotimes_{i=1}^{n} \begin{pmatrix} \cos(\pi x_i/2) \\ \sin(\pi x_i/2) \end{pmatrix}.$$

Rotation matrices for $k \in \{x, y, z\}$: $R_k(\alpha) = e^{-i\alpha\sigma_k/2}$,

$$R_y(\alpha) = \begin{pmatrix} \cos\frac{\alpha}{2} & -\sin\frac{\alpha}{2} \\ \sin\frac{\alpha}{2} & \cos\frac{\alpha}{2} \end{pmatrix}, \quad R_z(\alpha) = \begin{pmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{pmatrix}.$$

- One feature per qubit; $n = d$. Depth $O(1)$.
- Product state — *no entanglement yet*; entanglement added by two-qubit gates in the variational layers.
- The output $f(x; \theta)$ is a *Fourier series* in x with frequencies determined by the generator spectrum.

Amplitude Encoding

Idea. Embed $\mathbf{x} \in \mathbb{R}^{2^n}$, $\|\mathbf{x}\| = 1$, into state amplitudes:

$$|\phi(\mathbf{x})\rangle = \sum_{j=0}^{2^n-1} x_j |j\rangle.$$

Capacity.

- n qubits encode 2^n features — *exponential compression*.
- Exploits the full Hilbert space.

Challenges.

- General state preparation requires $O(2^n)$ gates.
- Normalisation: $\sum_j x_j^2 = 1$.
- N data points need N circuit runs.

Read-out bottleneck

The exponential compression does *not* automatically give a speedup: reading out all 2^n amplitudes requires exponentially many measurements. This is the fundamental quantum read-out bottleneck relevant to all QPINNs.

Structured Feature Maps: IQP and ZZ

IQP feature map (Havlíček et al., Nature 2019):

$$|\phi(\mathbf{x})\rangle = U_\Phi(\mathbf{x})|+\rangle^{\otimes n}, \quad U_\Phi(\mathbf{x}) = e^{i\phi(\mathbf{x})}, \quad \phi(\mathbf{x}) = \sum_S c_S(\mathbf{x}) \bigotimes_{i\in S} Z_i.$$

Second-order ZZ feature map (two layers):

$$U_\Phi^{(2)}(\mathbf{x}) = V_2(\mathbf{x}) H^{\otimes n} V_1(\mathbf{x}) H^{\otimes n},$$
$$V_k(\mathbf{x}) = \exp\Big(i \sum_i x_i Z_i + i \sum_{i<j} (\pi - x_i)(\pi - x_j) Z_i Z_j \Big).$$

These encode two-body correlations; the resulting kernel $K(\mathbf{x}, \mathbf{x}') = \left|\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle\right|^2$ is believed to be classically hard to simulate. For QPINNs, IQP encoding introduces cross-frequency terms $\omega_i \omega_j$ that enrich the Fourier spectrum of the solution approximation.

Data Re-Uploading: Expressivity via Repetition

Key idea (Pérez-Salinas et al. 2020): encode $\mathbf{x}$ *at every layer*:

$$U(\mathbf{x}, \boldsymbol{\Theta}) = \prod_{\ell=1}^{L} \Big[W(\boldsymbol{\Theta}^{(\ell)})\, S(\mathbf{x}) \Big], \quad S(\mathbf{x}) = \bigotimes_i R_y(\pi x_i).$$

Why it helps.

- L re-uploads $\Rightarrow$ Fourier series up to freq. L.
- For PDEs: $\omega_{\max} = Ln/2$ (n qubits, L layers).

Fourier: $f(\mathbf{x}; \boldsymbol{\Theta}) = \sum_{\boldsymbol{\omega} \in \Omega} c_{\boldsymbol{\omega}}(\boldsymbol{\Theta}) e^{i\boldsymbol{\omega} \cdot \mathbf{x}}$. Frequencies fixed by circuit; coefficients by $\boldsymbol{\Theta}$.

Encoding comparison for PDEs

Encoding	**Qubits**	$\omega_{\max}$
Angle (1 layer)	n	$n/2$
Re-uploading	n	$Ln/2$
IQP / ZZ	n	$n^2/4$
Amplitude	n	$2^n/2$

Design rule

Choose n, L s.t. $Ln \geq 2\omega^\star$.

Variational Quantum Circuits: Setup

For a parameterised circuit $U(\boldsymbol{\theta}) = \prod_j G_j(\theta_j)$ with $G_j(\theta_j) = e^{-i\theta_j P_j/2}$, $P_j^2 = \mathbb{1}$:

$$f(\boldsymbol{\theta}) = \langle \psi_0 | U^\dagger(\boldsymbol{\theta})\, \hat{B}\, U(\boldsymbol{\theta}) | \psi_0 \rangle.$$

Parameter-shift rule: derivation

Write $U(\theta_j) = A_j G_j(\theta_j) B_j$, $G_j = \cos\frac{\theta_j}{2}\mathbb{1} - i\sin\frac{\theta_j}{2} P_j$:

$$\frac{\partial f}{\partial \theta_j} = \frac{1}{2}\big[f(\theta_j + \tfrac{\pi}{2}) - f(\theta_j - \tfrac{\pi}{2})\big].$$

Exact; 2 circuit evaluations per parameter. Generalises to $r[f(\theta_j + \frac{\pi}{4r}) - f(\theta_j - \frac{\pi}{4r})]$ for eigenvalues $\pm r$.

2nd deriv.:
$\partial^2_{\theta_j} f = f(\theta_j{+}\pi) - 2f(\theta_j) + f(\theta_j{-}\pi)$.

Used in QFI and Hessian optimisers; combined with input-shift gives mixed PDE derivatives.

Quantum Fisher Information Matrix

Definition (pure state $|\psi(\boldsymbol{\theta})\rangle$):

$$F_{jk}(\boldsymbol{\theta}) = 4\,\mathrm{Re}\big(\langle\partial_j\psi|\partial_k\psi\rangle - \langle\partial_j\psi|\psi\rangle\langle\psi|\partial_k\psi\rangle\big)$$

where $|\partial_j\psi\rangle = \partial|\psi(\boldsymbol{\theta})\rangle/\partial\theta_j$.

Geometric meaning. $\mathbf{F}$ is the **Fubini–Study metric** on the variational manifold:

$$ds^2 = \sum_{jk} F_{jk}\, d\theta_j\, d\theta_k.$$

It measures the *information-geometric distance* between neighbouring quantum states.

Covariance form. For product-of-rotations circuits:

$$F_{jk} = 4\,\mathrm{Cov}_\psi(G_j, G_k).$$

Circuit formula (2-fold shifts)

For overlap $\mathcal{O}(\boldsymbol{\theta}) = |\langle\psi_{\mathrm{ref}}|\psi\rangle|^2$:

$$F_{jk} = \tfrac{1}{2}[\mathcal{O}_{j+,k+} - \mathcal{O}_{j+,k-} - \mathcal{O}_{j-,k+} + \mathcal{O}_{j-,k-}].$$

Shift each index by $\pm\frac{\pi}{2}$. Cost: $O(p^2)$ circuit evals.

Natural Gradient and the TDVP

Natural gradient (Amari 1998; Stokes et al. 2020):

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, \mathbf{F}(\boldsymbol{\theta})^{+}\, \nabla_{\theta} C.$$

Derivation. Minimise $C(\boldsymbol{\theta} + \delta\boldsymbol{\theta})$ subject to fixed Bures distance: $\delta\boldsymbol{\theta}^{\top}\mathbf{F}\,\delta\boldsymbol{\theta} = \varepsilon^{2}$, giving $\delta\boldsymbol{\theta} \propto -\mathbf{F}^{-1}\nabla C$.
The update is *reparameterisation-invariant*: it always descends the steepest direction on $\mathcal{M}$.

TDVP equations of motion

From the Dirac–Frenkel principle, projecting $(i\partial_t - H)|\psi\rangle = 0$ onto the tangent space $\{|\partial_j\psi\rangle\}$:

$$\sum_k F_{jk}\,\dot{\theta}_k = 2\,\mathrm{Im}\langle\partial_j\psi|H|\psi\rangle$$

Imaginary-time TDVP = natural gradient VQE

Substituting $t \to -i\tau$:
$\dot{\boldsymbol{\theta}} = -\mathbf{F}^{-1}\nabla_{\theta} E(\boldsymbol{\theta})$. Equivalent to Stochastic Reconfiguration (SR) in VMC.

Effective Dimension and Expressivity

Abbas et al. (2021) define the **effective dimension** via the average log-volume of the QFI:

$$d_{\text{eff}} = 2\,\frac{1}{\log N}\log\left[\frac{1}{V}\int_{\boldsymbol{\theta}}\sqrt{\det\left(I + \frac{N}{2\pi\ln N}\,\mathbf{F}(\boldsymbol{\theta})\right)}\,d\boldsymbol{\theta}\right],$$

where N is the number of training samples.

- High $d_{\text{eff}} \Rightarrow$ model fits many functions; good generalisation capacity.
- Some QNNs achieve higher d_{eff} than classical networks of comparable parameter count.
- $\mathbf{F} \approx 0$ (rank-deficient) $\Rightarrow$ parameters are redundant; barren-plateau-like flat directions.

QFI as trainability indicator

$$\left|\frac{\partial C}{\partial \theta_j}\right|^2 \leq \lambda_{\max}(\mathbf{F}) \cdot \mathrm{Var}_{\psi}(\hat{B})$$

(Cauchy–Schwarz on the Fubini–Study metric).

Small $\lambda_{\max}(\mathbf{F}) \Leftrightarrow$ all gradients are small $\Leftrightarrow$ barren plateau.

Barren Plateaus: Definition and Origin

Definition (McClean et al. 2018)

$\{U(\boldsymbol{\theta})\}$ has a **barren plateau** if:

$$\mathbb{E}_{\boldsymbol{\theta}}\big[\partial_{\theta_j} C\big] = 0, \qquad \mathrm{Var}_{\boldsymbol{\theta}}\big[\partial_{\theta_j} C\big] = O(b^{-n}),\ b > 1.$$

Mathematical origin. For Haar-random U on $\mathrm{SU}(2^n)$:

$$\mathbb{E}_U[\langle \hat{O} \rangle] = \tfrac{\mathrm{Tr}\,\hat{O}}{2^n}, \quad \mathrm{Var}[\partial_{\theta_j} C] \sim \tfrac{\mathrm{Tr}\,\hat{O}^2}{4^n}.$$

For a local observable $\hat{O} = Z_k$: $\mathrm{Var} \sim 1/2^n$.

Concentration of measure (Lévy's lemma): in high dimensions, all Lipschitz functions concentrate near their mean with probability $\geq 1 - 2e^{-\Omega(2^{2n}\varepsilon^2)}$.

Consequence

Deep random circuits make $C(\boldsymbol{\theta}) \approx \mathrm{Tr}\,\hat{O}/2^n$ almost everywhere: landscape is flat. This is *not* a local-minima problem.

Noise-Induced Barren Plateaus and Mitigation

Hardware noise (Wang et al. 2021):
depolarising noise with rate ϵ per gate gives:

$$|\langle \hat{O} \rangle_{\text{noisy}}| \leq (1-\epsilon)^{pd} |\langle \hat{O} \rangle_{\text{ideal}}|,$$

and $\text{Var}[\partial_{\theta_j} C_{\text{noisy}}] \sim (1-\epsilon)^{2pd} \text{Var}[\partial_{\theta_j} C_{\text{ideal}}]$.
Affects even *shallow* circuits with many gates.

Mitigation strategies

Strategy	Mechanism
Local cost	poly(n) scaling
Problem ansatz	exploit symmetries
Layerwise training	$O(1)$ depth locally
Identity init	$O(1)$ init gradients
Natural gradient	QFI-respecting steps
Shallow circuits	avoid Haar regime

QPINNs and BPs

Physics-inspired shallow ansätze naturally avoid the random circuit regime — PDE structure guides the ansatz design.

Entanglement, QNTK, and Training Regimes

Entanglement entropy:
$S(\rho_A) = -\operatorname{Tr}(\rho_A \log \rho_A)$ governs the training regime:

Regime	$S(\rho_A)$	Training
Low	≈ 0	Classically simulable
Volume-law	$\Theta(n)$	Barren plateaus
Area-law	$O(\lvert\partial A\rvert)$	Trainable

Quantum Neural Tangent Kernel (QNTK):

$$\Theta(\mathbf{x}, \mathbf{x}') = \sum_j \frac{\partial f(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_j} \frac{\partial f(\mathbf{x}';\boldsymbol{\theta})}{\partial \theta_j}.$$

- Governs training dynamics in the linearised regime.
- At a barren plateau: $\Theta \approx 0$ for all $\mathbf{x}, \mathbf{x}'$.
- Bound: $\Theta(\mathbf{x}, \mathbf{x}) \leq \lambda_{\max}(\mathbf{F})$.
- Connects QNN training to quantum kernel methods (QSVMs).

Classical PINNs: Raissi, Perdikaris, Karniadakis (2019)

A classical PINN approximates the solution of $\mathcal{N}[u](x,t)=0$ on Ω by a neural network $u_\theta(x,t)$ minimising:

$$\mathcal{L}(\theta) = \underbrace{\tfrac{1}{N_r}\sum_k |\mathcal{N}[u_\theta](x_k,t_k)|^2}_{\mathcal{L}_r} + \lambda \underbrace{\tfrac{1}{N_b}\sum_k |u_\theta(x_k^b,t_k^b)-g_k|^2}_{\mathcal{L}_b}$$

Collocation points $\{x_k\}$ sampled in Ω; $\{x_k^b\}$ on $\partial\Omega$.

Automatic differentiation

Classical PINNs use autograd (no finite differences).

Spectral bias

Classical NNs learn low-frequency components much faster than high-frequency ones — motivation for QPINNs with explicit frequency spectra.

Examples

Poisson $-\nabla^2 u = f$; Heat $\partial_t u = \kappa\nabla^2 u$; Wave $\partial_{tt} u = c^2\nabla^2 u$; Burgers $\partial_t u + u\partial_x u = \nu\partial_{xx} u$; Schrödinger $i\partial_t\psi = H\psi$.

PINN Loss Landscape and the Neural Tangent Kernel

Residual at collocation point x_k:
$r_k(\theta) = \mathcal{D}[f_\theta](x_k)$.

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_k r_k^2, \quad \nabla_\theta \mathcal{L} = \frac{2}{N}\sum_k r_k \, \nabla_\theta r_k.$$

The **Neural Tangent Kernel (NTK)** governs training dynamics:

$$\Theta_{kl} = \nabla_\theta f_\theta(x_k)^\top \nabla_\theta f_\theta(x_l).$$

Eigenvalues of Θ determine convergence rates per frequency component.

Failure modes

1. **Spectral bias:** high-frequency modes converge slowly
2. **Loss balance:** $\mathcal{L}_r$ and $\mathcal{L}_b$ compete
3. **Stiff equations:** chaotic landscape
4. **High dimensionality:** exponential cost

Quantum motivation

QPINNs address points 1 and 4: richer frequency spectra via entanglement; quantum memory scales as $O(2^n)$ with n qubits.

QPINN: The Quantum Ansatz for PDE Solutions

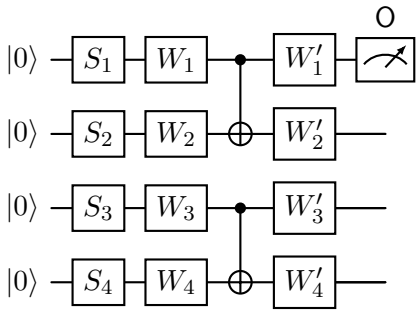
Replace the classical network $u_\theta(x)$ with a variational quantum circuit output:

$$u_\theta(x) = \langle 0|U^\dagger(x,\theta)\, O\, U(x,\theta)|0\rangle$$

where:

- $U(x,\theta) = W_L(\theta)\cdots W_1(\theta)\, S(x)$
- $S(x) = \bigotimes_i R_y(\pi x)$: data encoding
- $W_l(\theta)$: variational Pauli layers
- $O = Z_0$: fixed observable giving scalar output

The output $u_\theta(x) \in [-1, 1]$ approximates the normalised PDE solution.



S_i: encoding; W_i: variational; $O = Z_0$.

The QPINN Loss Function

For $\mathcal{D}[u](x) = f(x)$ on Ω with $u = g$ on $\partial\Omega$:

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N_r}\sum_k \left|\mathcal{D}[u_\theta](x_k) - f(x_k)\right|^2}_{\mathcal{L}_r} + \lambda \underbrace{\frac{1}{N_b}\sum_k \left|u_\theta(x_k^b) - g(x_k^b)\right|^2}_{\mathcal{L}_b}.$$

Gradient w.r.t. θ_μ:

$$\frac{\partial \mathcal{L}_r}{\partial \theta_\mu} = \frac{2}{N_r}\sum_k r_k \frac{\partial \mathcal{D}[u_\theta](x_k)}{\partial \theta_\mu}.$$

For a *linear* operator $\mathcal{D}$:

$$\frac{\partial \mathcal{D}[u_\theta]}{\partial \theta_\mu} = \mathcal{D}\left[\frac{\partial u_\theta}{\partial \theta_\mu}\right],$$

so $\mathcal{D}$ and the parameter-shift commute.

Two shift rules working together

- **Input-shift**: computes spatial derivatives $\mathcal{D}[u_\theta]$
- **Parameter-shift**: computes gradients $\partial u_\theta / \partial \theta_\mu$

Total cost per gradient step: $O(N_p \times N_r)$ circuit evaluations.

The Input-Shift Rule: Spatial Derivatives

For angle encoding $S(x) = e^{-ixP/2}$ (Pauli P, eigenvalues $\pm\frac{1}{2}$):

First derivative (input-shift rule):

$$\frac{\partial u_\theta}{\partial x} = \tfrac{1}{2}\left[u_\theta(x+\tfrac{\pi}{2}) - u_\theta(x-\tfrac{\pi}{2})\right]$$

Second derivative:

$$\frac{\partial^2 u_\theta}{\partial x^2} = u_\theta(x+\tfrac{\pi}{2}) - 2u_\theta(x) + u_\theta(x-\tfrac{\pi}{2})$$

n-th order:

$$\partial_x^n u_\theta = \sum_{k=0}^{n} (-1)^k \binom{n}{k} u_\theta\Big(x + \tfrac{(n-2k)\pi}{4}\Big)$$

Key result

All spatial derivatives of $u_\theta(x)$ are computable by evaluating the *same circuit* at shifted input values. No ancilla qubits; no additional circuit depth.

Cost budget

k-th order derivative at one point: $k+1$ circuit runs.

Laplacian ∂_{xx}: **3 runs** at $x-\frac{\pi}{2}$, x, $x+\frac{\pi}{2}$.

Unified Shift Framework: One Rule, Two Applications

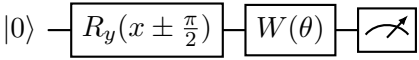
For *any* gate $e^{-i\phi P/2}$ (Pauli P):

$$\frac{\partial}{\partial\phi}\langle O\rangle_\phi = \frac{1}{2}\left[\langle O\rangle_{\phi+\pi/2} - \langle O\rangle_{\phi-\pi/2}\right].$$

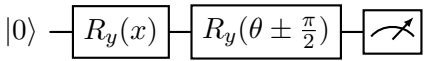
Two interpretations:

- $\phi = \theta_\mu$ (variational parameter): **parameter-shift** — gradient for training
- $\phi = x$ (input encoding): **input-shift** — spatial derivative of PDE solution

They are *the same formula*, applied to different gates in the circuit.



Input-shift: $\partial_x u_\theta$.



Parameter-shift: $\partial_\theta u_\theta$.

Commutativity (linear PDEs)

The shifts commute: $\partial_{\theta_\mu}\mathcal{D}[u_\theta](x) = \mathcal{D}[\partial_{\theta_\mu} u_\theta](x)$. This is why QPINN training is tractable.

Multi-Dimensional PDEs and the Variational Form

Multi-dimensional encoding. For $u(x_1, \ldots, x_d)$ with n_i qubits for x_i:

$$S(\mathbf{x}) = \bigotimes_{i=1}^{d} R_y(\pi x_i)^{\otimes n_i}.$$

Cross-derivatives: shift simultaneously in x_i and x_j: $\partial_{x_i}\partial_{x_j} u_\theta$ via simultaneous input-shifts. Total qubits: $n = n_1 + \cdots + n_d$. Classical PINN: $O(\varepsilon^{-d})$ parameters; QPINN: $O(d)$ qubits if solution has tensor-product structure.

Variational (Ritz) form

For Poisson $-\nabla^2 u = f$, minimise:

$$\mathcal{L}_{\text{Ritz}} = \sum_k w_k \left[\tfrac{1}{2}(u'_\theta(x_k))^2 - f_k u_k\right],$$

Only first derivatives needed; smoother landscape.

Ritz-QPINN = quantum Galerkin

The ansatz $u_\theta(x)$ plays the finite element basis role but spans an exponentially large function space.

The 1D Poisson Equation

Problem:

$$-\frac{d^2u}{dx^2} = f(x), \quad x \in (0,1), \quad u(0) = u(1) = 0.$$

Exact solution for $f(x) = \pi^2 \sin(\pi x)$:

$$u^\star(x) = \sin(\pi x).$$

QPINN residual loss:

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{k=1}^{N_r} \left[-\frac{d^2 u_\theta}{dx^2}\bigg|_{x_k} - f(x_k) \right]^2.$$

Boundary loss: $\mathcal{L}_b = u_\theta(0)^2 + u_\theta(1)^2$.

Second derivative via input-shift:

$$\frac{d^2 u_\theta}{dx^2}\bigg|_x = u_\theta(x+\tfrac{\pi}{2}) - 2u_\theta(x) + u_\theta(x-\tfrac{\pi}{2}).$$

This uses $S(x) = e^{-i\pi x P/2}$ with $P = Y \otimes \cdots$: exactly 3 circuit evaluations per collocation point.

Circuit count per gradient step

N_r interior points $\times$ 3 (Laplacian) $\times$ $2N_p$ (parameter shift) $= 6N_pN_r$ total. For $N_p = 8$, $N_r = 20$: 960 evaluations per step.

Python: Circuit and Gate Definitions

```python
import numpy as np

def ry(t):
    c, s = np.cos(t / 2), np.sin(t / 2)
    return np.array([[c, -s], [s, c]], dtype=complex)

I2   = np.eye(2, dtype=complex)
CNOT = np.array([[1,0,0,0],[0,1,0,0],[0,0,0,1],[0,0,1,0]], dtype=complex)
Z0   = np.kron(np.diag([1., -1.]).astype(complex), I2)  # Z on qubit 0

def var_layer(p):
    """Ry(p0)xRy(p1) -> CNOT -> Ry(p2)xRy(p3)"""
    return CNOT @ np.kron(ry(p[0]), ry(p[1]))

def circuit(x, params):
    """
    2-qubit QPINN circuit for 1D PDE.
    Encoding: Ry(pi*x) on each qubit.
    Ansatz: 2 variational layers (8 parameters).
    Output: <Z_0> in [-1, 1].
    """
    state = np.zeros(4, dtype=complex); state[0] = 1.0
    U = np.kron(ry(np.pi * x), ry(np.pi * x))       # angle encoding
    U = var_layer(params[:4]) @ U                    # layer 1
    U = var_layer(params[4:8]) @ U                   # layer 2
    psi = U @ state
    return float((psi.conj() @ Z0 @ psi).real)
```

Python: Input-Shift Derivatives and Loss

```
SHIFT = np.pi / 2

def du_dx(x, params):
    return 0.5 * (circuit(x + SHIFT, params)
                - circuit(x - SHIFT, params))

def d2u_dx2(x, params):
    return (circuit(x + SHIFT, params)
          - 2 * circuit(x, params)
          + circuit(x - SHIFT, params))

x_r, x_b, lambda_bc = np.linspace(0.05,0.95,20), np.array([0.,1.]), 10.

def loss(params):
    res = [-d2u_dx2(x,params) - np.pi**2*np.sin(np.pi*x) for x in x_r]
    L_r = np.mean(np.array(res)**2)
    L_b = circuit(x_b[0],params)**2 + circuit(x_b[1],params)**2
    return L_r + lambda_bc * L_b
```

Python: Gradient via Parameter-Shift Rule

```python
def grad_loss(params, shift=np.pi / 2):
    """Exact gradient via parameter-shift rule."""
    g = np.zeros(len(params))
    for mu in range(len(params)):
        p_plus  = params.copy(); p_plus[mu]  += shift
        p_minus = params.copy(); p_minus[mu] -= shift
        g[mu] = 0.5 * (loss(p_plus) - loss(p_minus))
    return g
```

Python: Training Loop (Adam Optimiser)

```python
np.random.seed(7)
params = np.random.uniform(0, np.pi, 8)    # 2 layers x 4 params

# Adam hyperparameters
lr, b1, b2, eps = 0.15, 0.9, 0.999, 1e-8
m, v = np.zeros(8), np.zeros(8)
history = []

for step in range(300):
    g = grad_loss(params)
    t = step + 1
    m  = b1 * m + (1 - b1) * g
    v  = b2 * v + (1 - b2) * g**2
    mc = m / (1 - b1**t);  vc = v / (1 - b2**t)
    params -= lr * mc / (np.sqrt(vc) + eps)
    history.append(loss(params))
    if (step + 1) % 75 == 0:
        print(f"Step {step+1:4d}  loss = {history[-1]:.6f}")

# Evaluate and compare
x_test  = np.linspace(0, 1, 100)
u_qpinn = np.array([circuit(x, params) for x in x_test])
u_exact = np.sin(np.pi * x_test)
l2_err  = np.sqrt(np.mean((u_qpinn - u_exact)**2))
print(f"\nL2 error vs. sin(pi*x): {l2_err:.4f}")
# Expected: L2 error ~ 0.03 -- 0.05 after 300 Adam steps
```

Results and Interpretation

Expected output:

- L2 error $\lesssim 0.05$ after 300 steps
- Loss drops 2–3 orders of magnitude
- QPINN traces $\sin(\pi x)$ accurately

Improving accuracy:

- More qubits (wider frequency spectrum)
- More variational layers (richer coefficients)
- More collocation points N_r
- Natural gradient (QFI) instead of Adam

Connection to Section 2

Adam uses Euclidean geometry. Natural gradient $\delta\theta = -\eta \mathbf{F}^{-1} \nabla_\theta \mathcal{L}$ respects the Fubini–Study metric and converges faster on stiff, anisotropic loss landscapes.

Barren plateau warning

Adding more layers improves expressivity but risks entering the barren plateau regime. Physics-inspired ansätze mitigate this: use $L \leq \log n$ layers with structured entanglement.

Expressivity of QPINNs: Fourier Analysis

Theorem (Schuld et al. 2021). The output of any QNN with data-encoding gates $e^{-ixP/2}$ is a *partial Fourier series*:

$$u_\theta(x) = \sum_{\omega \in \Omega} c_\omega(\theta)\, e^{i\omega x}$$

where $\Omega \subseteq \{\omega_1 + \cdots + \omega_n : \omega_i \in \{0, \pm\frac{1}{2}\}\}$. Frequencies fixed by encoding; c_ω tuned by θ.
For PDEs: highest frequency in Ω limits the shortest spatial scale resolved.

Error decomposition

$$\|u_{\theta^\star} - u^\star\|_{L^2} \le \varepsilon_{\text{approx}} + \varepsilon_{\text{opt}} + \varepsilon_{\text{stat}}$$

Expressivity: $\varepsilon_{\text{approx}} \sim \sum_{\omega \notin \Omega} |\hat{u}^\star_\omega|$;
sampling: $\varepsilon_{\text{stat}} \sim N_r^{-1/2}$.

Circuit sizing

For target frequency $\omega^\star$, choose n qubits and L re-uploading layers such that $Ln \ge 2\omega^\star$.

Extension: Time-Dependent PDEs

For $u(x,t)$, encode space and time jointly:

$$S(x,t) = R_y(\pi x)^{\otimes n_x} \otimes R_y(\pi t)^{\otimes n_t}.$$

1D heat equation: $\partial_t u - \kappa \partial_{xx} u = 0$
Input shifts for PDE residual:

$$\partial_t u \approx \tfrac{1}{2}[u(x,t+\tfrac{\pi}{2}) - u(x,t-\tfrac{\pi}{2})]$$
$$\partial_{xx} u \approx u(x+\tfrac{\pi}{2},t) - 2u(x,t) + u(x-\tfrac{\pi}{2},t)$$

Each residual point: 5 circuit evaluations.
Add IC loss:
$\mathcal{L}_{\text{IC}} = N_x^{-1} \sum_k |u_\theta(x_k,0) - u_0(x_k)|^2$.

Dimensional scaling

d-dimensional PDE: $n_x + n_t = n$ total qubits. Classical PINN cost: $O(\varepsilon^{-d})$. QPINN cost: $O(d)$ qubits if solution has tensor-product structure. Advantage largest when $d \gg 1$.

Exact: $u(x,t) = e^{-\pi^2 t}\sin(\pi x)$

Compare QPINN to exact solution; measure L^∞ error across the space-time domain.

The Schrödinger Equation as a QPINN

Time-dependent Schrödinger equation:

$$i\hbar\partial_t\psi = H\psi, \quad H = -\frac{\hbar^2}{2m}\partial_{xx} + V(x).$$

Split $\psi = \psi_R + i\psi_I$:

$$\partial_t\psi_R = \tfrac{\hbar}{2m}\partial_{xx}\psi_I - V\psi_I/\hbar$$
$$\partial_t\psi_I = -\tfrac{\hbar}{2m}\partial_{xx}\psi_R + V\psi_R/\hbar$$

Two QPINNs: $\psi_R \approx \langle Z_0\rangle_{x,t,\theta_R}$, $\psi_I \approx \langle Z_0\rangle_{x,t,\theta_I}$.
All derivatives by input-shift; $V(x)$ analytically.

Bridge to Hamiltonian simulation

The QPINN Schrödinger ansatz is the *classical simulation* of quantum evolution. On hardware, $e^{-iHt/\hbar}$ is exact — the QPINN bridges PDE solving and digital quantum simulation.

Time-independent: VQE connection

$H\psi = E\psi$ solved by minimising $E[\psi_\theta] = \langle H\rangle_{\psi_\theta} \geq E_0$ (Rayleigh quotient = VQE objective). QFI/TDVP (Section 3) govern both VQE and QPINN.

Barren Plateaus in QPINNs: Additional Sources

QPINNs inherit the general barren plateau problem (Section 4) and have additional sources specific to PDE training:

General BP (random deep circuits):

$$\mathrm{Var}\left(\frac{\partial \mathcal{L}}{\partial \theta_\mu}\right) \sim \frac{1}{2^n}.$$

QPINN-specific sources:

- Residual $r_k \approx 0$ early in training: gradient is small even without BPs
- Loss imbalance: $\mathcal{L}_r \gg \mathcal{L}_b$ makes BC enforcement slow
- Stiff PDEs: widely varying eigenvalues of the differential operator

QPINN-specific mitigation

1. **Physics ansatz:** mirror Green's function structure
2. **Incremental training:** low modes first
3. **Adaptive collocation:** focus on large-residual points
4. **Loss weighting:** normalise $\mathcal{L}_r$, $\mathcal{L}_b$ by gradient norms
5. **Natural gradient:** $\dot{\theta} = F^{-1}\nabla_\theta \mathcal{L}$

QPINN vs. Classical PINN: Summary

	Classical PINN	**QPINN**
Ansatz	Deep MLP $u_\theta(x)$	Circuit $\langle O \rangle_{x,\theta}$
Spatial deriv.	Automatic differentiation	Input-shift rule (exact)
Parameter grad.	Backpropagation	Parameter-shift rule (exact)
Frequency	Arbitrary (spectral bias)	Bounded, explicit Ω
Memory	$O(N_p)$ params	$O(n)$ params, 2^n amplitudes
High-d PDE	Exponential cost	Potential $O(d)$ qubit advantage
Optimiser geom.	Euclidean	QFI / Fubini–Study (natural grad.)
Advantage	Production-ready	High-d, quantum PDEs

The central message

QPINN = physics-informed QNN: circuit $\approx$ PDE solution; spatial derivatives via input-shift rule; training minimises residual with exact quantum gradients; QFI/TDVP give the natural optimisation geometry.

Key Equations

Concept	Formula
Parameter-shift rule	$\frac{\partial f}{\partial \theta_j} = \frac{1}{2}[f(\theta_j + \frac{\pi}{2}) - f(\theta_j - \frac{\pi}{2})]$
Input-shift rule	$\frac{\partial u_\theta}{\partial x} = \frac{1}{2}[u_\theta(x + \frac{\pi}{2}) - u_\theta(x - \frac{\pi}{2})]$
Laplacian (2nd order)	$\partial_{xx} u_\theta = u_\theta(x + \frac{\pi}{2}) - 2u_\theta(x) + u_\theta(x - \frac{\pi}{2})$
QFI (pure state)	$F_{jk} = 4\,\mathrm{Re}[\langle \partial_j \psi \vert \partial_k \psi \rangle - \langle \partial_j \psi \vert \psi \rangle \langle \psi \vert \partial_k \psi \rangle]$
Natural gradient	$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, \mathbf{F}^{-1} \nabla C$
TDVP	$\sum_k F_{jk} \dot{\theta}_k = 2\,\mathrm{Im} \langle \partial_j \psi \vert H \vert \psi \rangle$
Barren plateau	$\mathrm{Var}[\partial C / \partial \theta_j] \sim 1/2^n$
QPINN loss	$\mathcal{L} = \Vert \mathcal{D}[u_\theta] - f \Vert^2 + \lambda \Vert u_\theta - g \Vert^2_{\partial\Omega}$

Exercises

1. **Parameter-shift by hand.** For $f(\theta) = \langle 0|R_y(\theta)^\dagger Z R_y(\theta)|0\rangle$, apply the parameter-shift rule and verify by direct differentiation.
2. **QFI of a single qubit.** For $|\psi(\theta)\rangle = R_y(\theta)|0\rangle$, compute F_{11} via the Fubini–Study formula and interpret geometrically.
3. **Barren plateau scaling.** Numerically estimate $\mathrm{Var}[\partial C/\partial\theta_j]$ for $n = 2, \ldots, 8$ qubit random circuits of depth $d = 2n$. Verify the $O(1/2^n)$ scaling.
4. **Harmonic oscillator QPINN.** Modify the Poisson code to solve $u'' + \omega^2 u = 0$, $u(0) = 0$, $u'(0) = 1$. Compare to $u(x) = \sin(\omega x)/\omega$.
5. **Second-order input-shift identity.** Prove that for $S(x) = e^{-ixP/2}$ with eigenvalues $\pm\frac{1}{2}$:
$$\frac{d^2 u_\theta}{dx^2} = u_\theta(x+\tfrac{\pi}{2}) - 2u_\theta(x) + u_\theta(x-\tfrac{\pi}{2}).$$
6. **Natural gradient QPINN.** Replace Adam with the diagonal QFI natural gradient in the Poisson solver and compare convergence.

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